

LECTURE 2.2



# Goodness of Fit

ECONXXXXX Econometrics I

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PLACEHOLDER TERM

# Where We Are

## 1. Foundations

1.1 Data and Variation

1.2 Conditional Means

1.3 Sampling Variation

## 2. Simple Regression

2.1 Fitting a Line

**2.2 Goodness of Fit**

2.3 Bias and Precision

2.4 Reporting an Estimate

## 3. Multiple Regression

3.1 Adding Controls

3.2 Functional Form

3.3 Joint Tests

## 4. Limits

4.1 Diagnosing Violations

4.2 Limits of Regression

### Block 2 Is Four Lectures on One Estimator

Lectures already given are greyed and today's is marked. Lecture 2.1 fitted the line. Today asks how much of the variation in earnings it accounts for, 2.3 asks how far it lands from the truth, and 2.4 asks how to report it.

## How Small Is a Small Sum of Squared Residuals?

Lecture 2.1 chose the line that makes the sum of squared residuals smaller than any other line does, and on our 120 people that total is 2,692.8.

That number is in squared euro per hour, and it grows with the number of people in the sample. It has no upper limit and no natural scale, so on its own it ranks lines and nothing else.

Today builds the comparison it is missing: how much earnings vary in this sample before schooling is mentioned at all.

### Question

If I tell you the sum of squared residuals for a regression is 2,692.8, have I told you anything about how well the line fits?

# Lecture Plan

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## 1 Fitted Values and Residuals.

Splitting one person's earnings into the part the line gets and the part it misses.

## 2 Sums of Squares.

Adding those parts over 120 people, and deriving the identity that links the three totals.

## 3 What R-Squared Will Not Tell You.

Why a share of variation is not a verdict on a model or on a slope.

## SECTION 1

# Fitted Values and Residuals

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1. Fitted Values and Residuals
2. Sums of Squares
3. What R-Squared Will Not Tell You

# Splitting One Person's Earnings Into Two Parts

The fitted line assigns every person a predicted wage, and whatever is left over belongs to that person alone. Writing one person as the sum of the two is the whole of this section (Wooldridge, 2020):

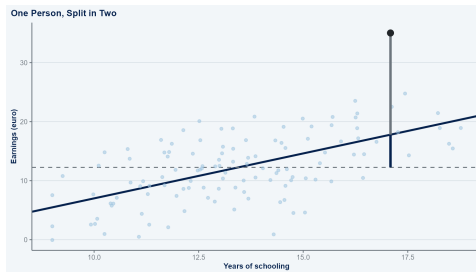
$$y_i = \hat{y}_i + \hat{u}_i \quad \text{where} \quad \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1(x_i - 13) \quad (1)$$

- $y_i$ : what person  $i$  actually earns per hour
- $\hat{y}_i$ : the fitted value, what the line predicts for person  $i$
- $\hat{u}_i$ : the residual, the amount the line misses person  $i$  by
- $\bar{y}$ : the average of  $y$  over the 120 people, 12.2490

## A Residual Is Not an Error

$u_i$  is the population error from 2.1, and we never see it.  $\hat{u}_i$  is a number we computed from the fitted line, so it inherits every flaw in that line. If we had fitted a different line, every residual would be different while every error stayed exactly as it was.

# The Split Shown on One Person



**Figure 1:** One person's earnings split into a fitted value and a residual.

## Three Heights, Two of Which We Choose

The dashed horizontal line is  $\bar{y}$ , at 12.2490. The solid line is the fitted line from 2.1, slope 1.5227 and intercept 11.5680. Person  $i$  sits above both, and the distance from  $\bar{y}$  up to that person is the sum of two segments:  $\bar{y}$  up to the line, and the line up to the person.

## Three Facts About the Residuals, From Lecture 2.1

Least squares did not just pick a line. It picked the pair that makes these two sums exactly zero, and every step today rests on them (Wooldridge, 2020):

$$\sum_{i=1}^n \hat{u}_i = 0 \qquad \sum_{i=1}^n (x_i - 13) \hat{u}_i = 0 \qquad (2)$$

The first says the residuals average to zero, so the fitted values average to  $\bar{y}$  as well. The second says the residuals retain no co-movement with schooling. Together they put the point  $(\bar{x}, \bar{y})$  on the fitted line.

### These Two Sums Are the Only Inputs to Today's Derivation

Neither holds because of an assumption about the population. Both hold by construction, for any sample of any 120 people, because the line was chosen to make them hold.

## A Large Residual Is Information About the Model

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In 2.1 the largest residual was 17.23, and it supplied 296.9 of the total 2,692.8. It is tempting to read that person as an error in the data.

They are not. A residual is a statement about the pair, the person and the line, and it cannot be assigned to one of them from its size alone. A person who earns far more than thirteen years of schooling predicts may have a rare occupation, a mistyped wage, or a feature the line was never given.

Lecture 4.1 is about reading residuals systematically: which patterns in them indicate a fault in the specification rather than a fact about a person.

### **The Residuals Sum to Zero Whatever Is Wrong With the Line**

The two conditions above hold for a badly specified line exactly as they hold for a well specified one. So no residual total can tell you the model is right, and today's R-squared inherits that limitation in full.

## SECTION 2

# Sums of Squares

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1. Fitted Values and Residuals
2. Sums of Squares
3. What R-Squared Will Not Tell You

## Three Totals, One for Each Kind of Variation

The per-person split becomes a comparison only once it is added over everybody. Three totals do that, and each answers one question (Wooldridge, 2020):

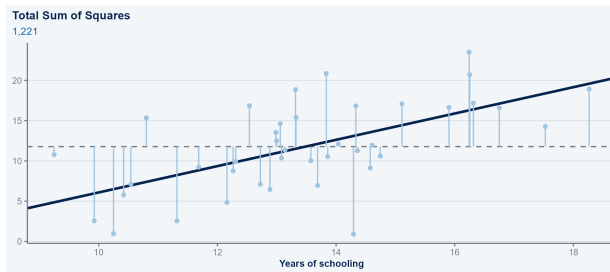
$$\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2 \quad \text{ESS} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 \quad \text{RSS} = \sum_{i=1}^n \hat{u}_i^2 \quad (3)$$

- TSS: how much earnings vary, before schooling is mentioned
- ESS: how much the fitted values vary around the same average
- RSS: how much is left once the line has been subtracted

### Wooldridge Writes SST, SSE and SSR for These Three

The same three totals, different letters. Wooldridge notes that the first is called SST or TSS with little confusion, but that SSE reads as “error sum of squares” and is not: it is the explained total (Wooldridge, 2020). Read the letters in whichever book is open and match them to the formula, not the name.

# The Total Sum of Squares as Vertical Segments

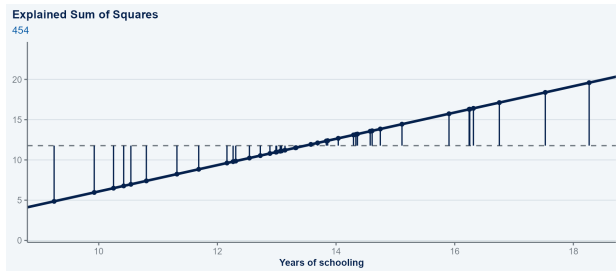


**Figure 2:** Each segment runs from a person to the sample average of earnings.

## The Benchmark Is the Average, Not the Line

This is the total we would face if we had never collected schooling: predict 12.2490 for everybody and square what is left. On our 120 people the segments square and sum to 4,162.6. Every later total is measured against this one.

# The Explained Sum of Squares as Vertical Segments



**Figure 3:** Each segment now runs from the fitted line down to the sample average.

## A Flat Line Would Contribute Nothing Here

These segments record how far the line departs from predicting the average for everybody. If  $\hat{\beta}_1$  were zero, every fitted value would equal 12.2490 and this total would be zero. On our 120 it is 1,469.8. The residual segments are the ones drawn in 2.1, and they square and sum to 2,692.8.

## Deriving the Identity: Splitting the Deviation From the Mean

Take one person's distance from the average and add and subtract the fitted value, which splits it into the two segments on the last two exhibits:

$$y_i - \bar{y} = \hat{u}_i + (\hat{y}_i - \bar{y}) \quad (4)$$

Square both sides and sum over the 120 people. The square of a sum is three terms, and two of them are totals we have already named:

$$\underbrace{\sum_{i=1}^n (y_i - \bar{y})^2}_{\text{TSS}} = \underbrace{\sum_{i=1}^n \hat{u}_i^2}_{\text{RSS}} + 2 \sum_{i=1}^n \hat{u}_i (\hat{y}_i - \bar{y}) + \underbrace{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}_{\text{ESS}} \quad (5)$$

### The Identity Holds Only If the Middle Term Is Zero

Nothing so far is special to least squares: this much is true of any line at all. The whole argument reduces to one cross term, and the next frame is the only place where the line being the least-squares line matters.

## Deriving the Identity: Why the Cross Term Is Zero

Substitute the intercept 2.1 derived,  $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1(\bar{x} - 13)$ , into the fitted value. The centring constant cancels and what remains is a deviation from the sample mean of schooling:

$$\hat{y}_i - \bar{y} = \hat{\beta}_0 + \hat{\beta}_1(x_i - 13) - \bar{y} = \hat{\beta}_1(x_i - \bar{x}) \quad (6)$$

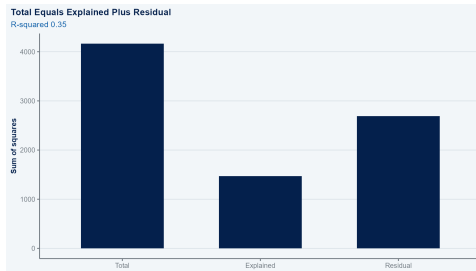
Put that into the cross term and take the constant  $\hat{\beta}_1$  outside:

$$\sum_{i=1}^n \hat{u}_i(\hat{y}_i - \bar{y}) = \hat{\beta}_1 \sum_{i=1}^n (x_i - \bar{x}) \hat{u}_i = \hat{\beta}_1 \left[ \sum_{i=1}^n (x_i - 13) \hat{u}_i - (\bar{x} - 13) \sum_{i=1}^n \hat{u}_i \right] = 0 \quad (7)$$

### Both Bracketed Sums Are Conditions From Lecture 2.1

The first is zero because the residuals retain no co-movement with schooling. The second is zero because the residuals sum to zero. Wooldridge reaches the same step at eq (2.37) (Wooldridge, 2020). R-squared behaves like a share only because both of 2.1's conditions hold.

# The Three Totals on Our 120 People



**Figure 4:** The three totals as bars: the first is the height of the other two.

## The Arithmetic Closes Exactly

$1,469.8 + 2,692.8 = 4,162.6$ . The explained total and the residual total add to the total variation with nothing left over, which is what the cross term vanishing bought us. Most of the height of the first bar is residual.

## Defining R-Squared From the Three Totals

The three totals are in squared euro per hour, so none of them can be compared across datasets. Dividing one by another removes the units and the sample size (Wooldridge, 2020):

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} \quad (8)$$

- $R^2$ : the share of the sample variation in earnings the line accounts for

### **R-Squared Is a Share Because the Identity Holds**

The two expressions agree only because ESS and RSS add to TSS, and  $R^2$  lies between nought and one for the same reason. Without the cross term vanishing it would be a ratio of two unrelated totals and could take any value at all.

## R-Squared on Our 120 People

Both routes to the number use the totals from the bar chart, and they agree because the identity holds:

$$R^2 = \frac{1469.8}{4162.6} = 0.3531 \qquad R^2 = 1 - \frac{2692.8}{4162.6} = 0.3531 \qquad (9)$$

### Reading the Number Out Loud

Schooling accounts for 35.31 per cent of the variation in hourly earnings across these 120 people. The other 64.69 per cent is variation the line does not reproduce. That sentence is about this sample and about the spread of earnings in it, and it says nothing yet about whether 1.5227 is a good estimate.

## R-Squared Is the Squared Correlation

Lecture 2.1 wrote the slope as a correlation times a ratio of standard deviations, which is Wooldridge's second form of eq (2.19) (Wooldridge, 2020):

$$\hat{\beta}_1 = \hat{\rho}_{xy} \cdot \frac{\hat{\sigma}_y}{\hat{\sigma}_x} \implies R^2 = \hat{\rho}_{xy}^2 \quad (10)$$

- $\hat{\rho}_{xy}$ : the sample correlation between schooling and earnings, 0.5942
- $\hat{\sigma}_y, \hat{\sigma}_x$ : the sample standard deviations of earnings and schooling

On our 120 people,  $0.5942^2 = 0.3531$ , which is the R-squared computed from the totals. The appendix does the three lines of substitution.

### In Simple Regression R-Squared Carries No New Information

With one regressor, R-squared is the correlation you already had, squared and stripped of its sign. It cannot tell you the slope is positive, and it is the same number whether earnings are regressed on schooling or schooling on earnings. In multiple regression it stops being a correlation, which is where its problems begin.

## Adding a Variable That Cannot Matter

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Two regressions on the same 120 people: earnings on schooling, and earnings on schooling together with each person's height in centimetres.

### **Question**

Which regression has the higher R-squared, and does your answer depend on whether height has anything to do with earnings?

## SECTION 3

# What R-Squared Will Not Tell You

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1. Fitted Values and Residuals
2. Sums of Squares
3. What R-Squared Will Not Tell You

## Adding Any Variable Raises the R-Squared

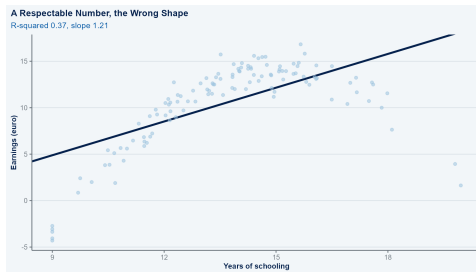
The height regression has the higher R-squared, and it does not depend on height mattering. Least squares minimises RSS over a larger set of candidate lines, so the new RSS cannot be larger than the old one. TSS is unchanged, since it never involved a regressor. So  $1 - \text{RSS}/\text{TSS}$  cannot fall.

A variable with no relation to earnings will still pick up some accidental co-movement in 120 people, so in practice the R-squared rises rather than staying still.

### **R-Squared Cannot Compare Specifications**

A criterion that never falls when a variable is added cannot choose between specifications, because it ranks the largest model first every time. Lecture 3.1 introduces adjusted R-squared, which charges for each variable; Lecture 3.3 does the comparison properly, with a test rather than a ranking.

# A Straight Line Through a Curved Relationship

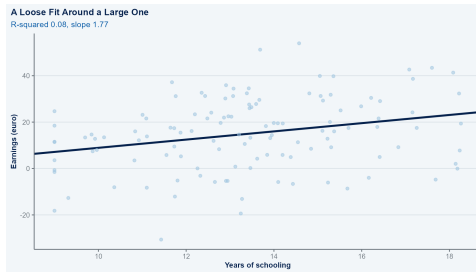


**Figure 5:** Earnings rise and then fall, while the fitted straight line rises throughout.

## The R-Squared Does Not Report the Curve

The straight line takes an average direction through a relation that reverses, and returns a single positive slope and a respectable R-squared. Nothing in either number records that the fit is wrong in shape. Lecture 3.2 fits the curve; the residual plot in Lecture 4.1 is how you find out you need to.

# A Low R-Squared Does Not Mean a Useless Slope



**Figure 6:** A wide scatter around a clearly rising line: little of the spread accounted for.

## Most of What Sets Earnings Is Not Schooling

Our own R-squared of 0.3531 says schooling leaves most of the spread in earnings unaccounted for, which is a fact about how varied Irish earnings are and not a fault in the line. Wooldridge's CEO salary regression returns 0.0132 on 209 firms, and he warns that a low R-squared does not make a regression useless (Wooldridge, 2020).

## Errors That Widen With Schooling Lower the R-Squared

Lecture 4.1 turns on heteroskedasticity: the errors stay centred on zero but spread out more at higher levels of schooling, while the true slope stays at 1.30. The fitted slope barely shifts and the R-squared falls to under a third of its value:

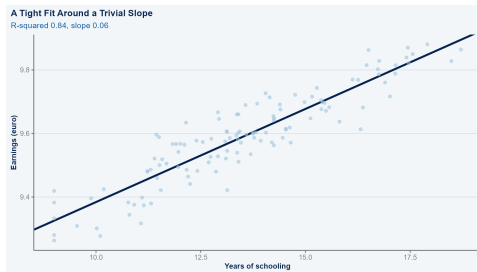
- baseline sample, errors of the same width at every schooling level: slope 1.5227,  $R^2 = 0.3531$
- heteroskedastic sample, errors that fan out with schooling: slope 1.4784,  $R^2 = 0.1014$

The slope answers “how much more does an extra year of schooling go with?” The R-squared answers “how tightly do people cluster around that answer?” Widening the errors at the top of the schooling range alters the second and leaves the first where it is.

### Two Different Questions, and Only One of Them Is About Size

R-squared has no units; the slope is in euro per hour per year of schooling. No value of a unit-free share can tell you whether a euro figure is large enough to matter, and Wooldridge notes that beginners weight R-squared far too heavily in judging a regression (Wooldridge, 2020).

# A High R-Squared Does Not Mean a Large Slope



**Figure 7:** A tight scatter around a nearly flat line: almost all the spread accounted for.

## The Slope Also Depends on the Two Standard Deviations

$\hat{\beta}_1 = \hat{\rho}_{xy} \cdot \hat{\sigma}_y / \hat{\sigma}_x$ , and R-squared fixes only the correlation. Holding the correlation at 0.99 and shrinking  $\hat{\sigma}_y$  shrinks the slope towards zero without touching the R-squared. Any R-squared is compatible with any slope, in either direction.

## Fit Measured Where the Line Was Chosen

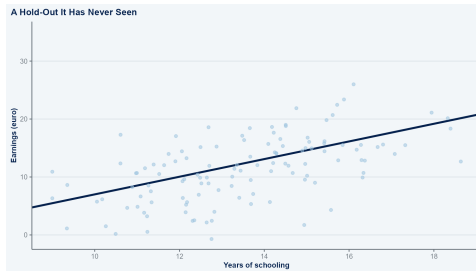
R-squared is computed on the same 120 people the coefficients were chosen to suit. Least squares picked the pair with the smallest possible RSS on those rows, so it collected every accident in them that a line can collect.

On a fresh 120 from the same population those accidents are different ones, and the line has no way to match them. Its RSS there will be larger, so the R-squared will be lower. The drop is not a failure; it is the part of the in-sample fit that was never about schooling.

### **The Second 120 Were Set Aside Before Anything Was Fitted**

Held out means held out. The moment a hold-out sample is used to choose a specification it stops being one, because the chosen line has then been suited to those rows too, and the honest comparison is gone.

# R-Squared on the Hold-Out Sample



**Figure 8:** The fitted line from our 120, drawn through 120 people it has never seen.

## 0.3531 In Sample, 0.2886 on the Hold-Out

The same line, the same population, the same slope of 1.5227: only the rows differ. A fall of 0.0645, about a fifth of the in-sample figure, is the optimism in fit measured on the data the line was chosen with. Report which of the two numbers you are quoting.

# What Would Have To Be True?

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This time the question is asked of the sentence we read out loud, that schooling accounts for 35.31 per cent of the variation in earnings in this sample.

## **Question**

What would have to be true for this number to mean what we just said it means? And what is the plausible story where it isn't?

## Application

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The app recomputes the three totals and the R-squared as you drag a single person up or down, so you can watch the slope and the R-squared respond by different amounts to the same movement.

### Question

Before we fit anything: will the R-squared computed on the hold-out 120 come out higher, lower, or about the same as the 0.3531 we got in sample?

The lab for this lecture computes R-squared by hand from the three totals, then checks it against `summary()`, and draws the residual plot.

# Summary: Residuals, Sums of Squares, and R-Squared

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## 1 Fitted Values and Residuals.

Every person splits into a fitted value and a residual, and the residuals sum to zero for any line least squares picks.

## 2 Sums of Squares.

$TSS = ESS + RSS$ , because the cross term vanishes under the two conditions from 2.1:  $1,469.8 + 2,692.8 = 4,162.6$ .

## 3 What R-Squared Will Not Tell You.

0.3531 is a share of spread, not a verdict: it cannot rank specifications, and it fell to 0.2886 on the hold-out.

## Next Lecture

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Lecture 2.3 asks a different question about the same line: not how well it fits these 120 people, but how far the slope lands from the 1.30 that generated them.

- **The Gauss-Markov Assumptions:** what must be true of the population first.
- **Unbiasedness:** the sense in which least squares gets the right answer.
- **Precision:** how far  $\hat{\beta}_1$  would move if we drew another 120.

The residual sum of squares returns there in a new role: 2,692.8 is what the standard error of the slope is built from.

## Before Lecture 2.3

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### 1 **Play:**

The app. Drag one person and watch the slope and the R-squared respond by different amounts.

### 2 **Apply:**

Problem set 2, covering 2.1 and 2.2. It repeats today's identity with  $n = 6$  and numbers you can add up by hand.

### 3 **Review:**

Wooldridge (2020), section 2-3, on fitted values, residuals, the three sums of squares and R-squared.

# Thank you

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**BACKUP**

# Appendix

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## Derivation: R-Squared Is the Squared Correlation

The body frame asserts  $R^2 = \hat{\rho}_{xy}^2$ . Start from the fitted deviation derived in the body,  $\hat{y}_i - \bar{y} = \hat{\beta}_1(x_i - \bar{x})$ , so the explained total is the slope squared times the schooling total:

$$R^2 = \frac{\hat{\beta}_1^2 \sum_{i=1}^n (x_i - \bar{x})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = \hat{\rho}_{xy}^2 \frac{\hat{\sigma}_y^2}{\hat{\sigma}_x^2} \cdot \frac{\hat{\sigma}_x^2}{\hat{\sigma}_y^2} = \hat{\rho}_{xy}^2 \quad (11)$$

### Dividing Both Totals by n Minus One Turns Them Into Variances

Substituting  $\hat{\beta}_1 = \hat{\rho}_{xy}\hat{\sigma}_y/\hat{\sigma}_x$  from p.26 gives the middle expression, and the two variance ratios cancel (Wooldridge, 2020). This is why R-squared carries no information beyond the correlation when there is one regressor.

## Notes and Tools

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- **The App:** the fit app, and the script that draws this deck's exhibits.
- **The Lecture in One Line:** `summary(lm(y ~ I(x - 13))$r.squared`.
- **This Lecture's Lab:** `fitted()`, `resid()`, and R-squared by hand from the three totals.

# References

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Wooldridge, J. M. (2020). *Introductory econometrics: A modern approach* (7th ed.). Cengage Learning.